SAP S/4HANA MM

Complete End-to-End Functional Configuration Documentation

Enterprise Structure · Master Data · P2P Cycle · Material Ledger · OBYC · Inventory · MRP · Special Procurement

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Düsseldorf Manufacturing AG (DMAG)

Fictional Sandbox Implementation Project

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Disclaimer: This portfolio is a fictional, sandbox-based SAP S/4HANA MM practice implementation created for learning and demonstration purposes. Düsseldorf Manufacturing AG is a fictional company used solely to illustrate SAP MM configuration concepts.

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PART 1

SAP S/4HANA MM Configuration Portfolio

Enterprise Structure · Master Data · Procurement · Valuation · Inventory

1. Company Background

Company: Düsseldorf Manufacturing AG (DMAG)

Industry: Precision Engineering · Automotive Components · Industrial Equipment Manufacturing

Headquarters: Düsseldorf, North Rhine-Westphalia, Germany

Company Code Currency: EUR | Group Currency: USD

Plants

Plant Code	Location	Purpose
DM01	Düsseldorf	Headquarters and Central Procurement
DM02	Dortmund	Production Plant
DM03	Stuttgart	Assembly and Quality Assurance

Storage Locations

1. SL01 — Raw Materials
2. SL02 — Semi-Finished Materials
3. SL03 — Finished Goods

Purchasing Organization

4. DMPO — Central Purchasing Organization (assigned to all three plants)
5. DMG — Procurement Specialist Purchasing Group

2. Enterprise Structure Configuration

The enterprise structure defines how DMAG is represented in SAP from company level down to storage location. All procurement and inventory transactions reference this structure.

Object	Code / Value	T-Code	Detail
Company	DMAG	OX15	Düsseldorf Manufacturing AG
Company Code	DM01	OX02	Currency: EUR (company) / USD (group) — OBY6
Plant 1	DM01	OX10	Düsseldorf — HQ and Procurement
Plant 2	DM02	OX10	Dortmund — Production
Plant 3	DM03	OX10	Stuttgart — Assembly and QA
Storage Loc	SL01–SL03	OX09	Created under each plant
Purchase Org	DMPO	OX08	Central — assigned to company code OX01 and all plants OX17
Purchase Group	DMG	OME4	Procurement Specialist group

SAP System Evidence *T-Code: OX10 / OX18 / OX17*	
Navigation: *SPRO → Enterprise Structure → Definition → Logistics General → Define Plant*	
Company Code:	`DM01 | Currency: EUR | Group Currency: USD`
Plants:	`DM01 Düsseldorf | DM02 Dortmund | DM03 Stuttgart`
Assignment:	`All three plants assigned to company code DM01 via OX18`

Purchase Org:	DMPO assigned to company code via OX01 and to all plants via OX17
Storage Locs:	SL01 Raw | SL02 Semi-Finished | SL03 Finished — created under each plant
What to verify: *Three plants visible in enterprise structure. DMPO linked to all plants. Storage locations are present under each plant.*	

3. Material Master Configuration

Material master is the central data repository for all materials. Configuration covers material types, number ranges, valuation classes and field selection.

3.1 Material Types and Valuation Classes

Material Type	Code	Valuation Class	Number Range	Sample Material
Raw Material	ROH	3000	200001–200999	200001 — Steel Sheet
Semi-Finished	HALB	7900	300001–300999	300001 — Machined Housing
Finished Goods	FERT	7920	400001–400999	400001 — Pump Assembly

3.2 Key Configuration Steps

Configuration	T-Code	Detail
Number Ranges	MMNR	Internal number ranges defined per material type
Field Selection	OMS9	Required, optional and display fields configured per screen per material type
Material Groups	OMSF	Groups for procurement categorization and spend reporting
Valuation Classes	OMSK	Valuation class assigned per material type — drives OBYC account determination
Delivery Tolerance	OME1	Purchasing Value Key: 10% over-delivery, 10% under-delivery

3.3 Material Master Views Activated

All materials extended to plants DM01, DM02 and DM03 with views: Basic Data 1 and 2, Classification, Purchasing, MRP 1–4, General Plant/Storage, Sales and Accounting.

SAP System Evidence *T-Code: MM01 / MM03*	
Navigation: *Logistics → Materials Management → Material Master → Material → Create (General)*	
Material:	200001 — Steel Sheet | Type: ROH | Base Unit: KG
Valuation Class:	3000 — Raw Material
Price Control:	V (Moving Average Price)
Purchasing View:	Purch Group: DMG | Delivery tolerance via purchasing value key
MRP View:	MRP Type: PD | Lot Size: EX | Safety Stock configured
Plants Extended:	DM01 Düsseldorf | DM02 Dortmund | DM03 Stuttgart
What to verify: *Material 200001 visible in MM03 with all views active. Extended to all three plants.*	

4. Business Partner — Vendor Master

In SAP S/4HANA, vendor data is managed through the Business Partner (BP) framework, replacing the legacy vendor master approach.

BP Role	Code	Purpose
General Data	0000	Name, address, communication — applies to all BP types
Supplier (Company Code)	FLVN00	Reconciliation account, payment terms, payment method
Supplier (Purchasing)	FLVN01	Purchase org data, order currency EUR, incoterms, GR-based IV flag

Account Group	Number Range	Vendor Type
L001 — Domestic Vendors	4000–4499	German and domestic EU suppliers
L002 — EU Vendors	4500–4999	International EU-based suppliers

SAP System Evidence *T-Code: BP*	
Navigation: *Accounting → Financial Accounting → Accounts Payable → Master Records → Maintain Centrally*	
BP Number:	`VEND-DM01 | Name: Steel Supplies GmbH`
Account Group:	`L001 — Domestic Vendor`
BP Roles:	`0000 General | FLVN00 Company Code | FLVN01 Purchasing`
Company Code:	`DM01 | Recon. Account: 160000 | Payment Terms: 30 days net`
Purchase Org:	`DMPO | Order Currency: EUR | GR-Based IV: Active`
What to verify: *All three BP roles assigned to VEND-DM01. Company code and purchasing org data visible.*	

5. Pricing Procedure — Condition Technique

Custom pricing procedure ZDM01 configured for DMAG using the four-step condition technique framework.

Step	Object	Code	T-Code	Configuration
1	**Condition Table**	700	M/03	Fields: Purch Org, Vendor, Material — basis for price lookup
2	**Access Sequence**	ZDM1	M/07	Single access step using condition table 700
3	**Condition Types**	PB00 / RA01 / FRB1	M/06	PB00 Gross Price | RA01 Discount % | FRB1 Freight
4	**Calculation Schema**	ZDM01	M/08	Gross Price → Discount → Freight → Net Value (NETWR)

Calculation Example — Steel Sheet 1,000 KG

Condition	Description	Rate	Amount (EUR)
PB00	Gross Price	2.50 EUR/KG	2,500.00
RA01	Volume Discount (5% for qty > 500 KG)	-5%	-125.00
FRB1	Freight Charges (fixed)	100 EUR flat	100.00
NETWR	Net Order Value	—	2,475.00

6. PO Release Strategy

Classification-based release strategy configured using four characteristics to determine approval levels for purchase orders.

6.1 Characteristics

Characteristic	SAP Field	Values
Plant	WERKS	DM01, DM02, DM03
Purchasing Organization	EKORG	DMPO
Document Type	BSART	NB (Standard PO), UB (STO), MK (Contract)
Net Order Value	GNETW	Value thresholds in EUR — four approval bands

6.2 Approval Levels

Net Order Value (EUR)	Required Approver	Release Code
0 to 50,000	Buyer	01
50,001 to 100,000	Buyer + Senior Buyer	01 + 02
100,001 to 200,000	Buyer + Senior Buyer + Procurement Manager	01 + 02 + 03
Above 200,000	All above + Finance Controller	01 + 02 + 03 + 04

Class: DM_CLASS_PORS | Class Type: 032 | Configured via CL02 (class) and CT04 (characteristics)

SAP System Evidence *T-Code: CL02 / CT04 / OMGSM*	
Navigation: *SPRO → Materials Management → Purchasing → Purchase Order → Release Procedure for Purchase Orders*	
Release Class:	`DM_CLASS_PORS | Class Type: 032`
Release Group:	`01 | Codes: 01 Buyer | 02 Sr Buyer | 03 Mgr. | 04 Finance`
Strategy Example:	`PO above 200,000 EUR — all four codes required before issue`
Workflow:	`ME29N for release | SAP Fiori Manage Workflows app also configured`
What to verify: *Release strategy active on document type NB. Test PO above 200,000 EUR shows blocked status pending all four approvals.*	

7. Material Ledger Configuration

Material Ledger is mandatory in SAP S/4HANA. Activated across all three DMAG plants with two parallel currencies.

Step	Activity	T-Code	DMAG Setting
1	**Check Currency Types**	FINSC_LEDGER	Leading ledger 0L: Type 10 EUR (Company Code), Type 30 USD (Group)
2	**Activate Actual Costing**	SPRO path (see note)	Actual costing active for DM01, DM02, DM03
3	**Actual Costing Run**	CKMLCP	Monthly period-end run: actual cost calculated and inventory revalued
4	**Material Price Analysis**	CKM3N	Review actual vs standard cost per material per period

8. OBYC — Automatic Account Determination

OBYC maps goods movements to FI GL accounts automatically. Every stock movement generates an accounting document based on the transaction key and valuation class.

Key	Description	Triggered By	GL Account — DMAG
BSX	Inventory Posting	GR (101), GI, STO, transfers	Val 3000→130000 Raw | Val 7920→131000 HALB | Val 7900→132000 FG

WRX	GR/IR Clearing	GR against PO (credit) | MIRO posting (debit)	191100 — GR/IR Clearing Account
PRD	Price Differences	Invoice price ≠ PO price (standard price materials)	281000 — Purchase Price Variance
GBB	Offsetting GI Entries	GI to production (261), cost center (201), scrap	GBB-VBR→400000 Consumption | GBB-VBO→401000
FRE	Freight Clearing	Freight condition posting at GR	410000 — Freight Clearing
KON	Consignment Payable	Withdrawal from consignment stock (201, 261)	160000 — Consignment Payable
INV	Inventory Differences	Physical inventory difference posting (MI07)	170000 — Inventory Difference

9. MRP Configuration

MRP generates procurement proposals automatically based on demand signals. MRP Live (MD01N) is the S/4HANA embedded engine.

MRP Parameter	DMAG Setting	T-Code
MRP Type	PD — Standard MRP (demand-based) | VB — Reorder Point	MM01 MRP 1 view
Lot Sizing	EX — Lot-for-Lot (exact shortfall quantity)	MM01 MRP 1 view
Safety Stock	Maintained on MRP 2 view — triggers PR before depletion	MM01 MRP 2 view
Plant Independent Req.	Demand entered across planning periods for HALB and FERT	MD61
MRP Live Execution	MD01N — S/4HANA embedded MRP, replaces classic MD01/MD02	MD01N
Results Review	Stock/requirements list — planned orders and PRs visible	MD04
Auto PO Conversion	Info Record and Source List prerequisites — ME59N batch conversion	ME01, ME11, ME59N

10. Special Procurement Scenarios

10.1 Stock Transport Order (STO) — DM02 to DM03

Step	Action	T-Code / Movement
1	**Create STO**	ME21N — document type UB, supplying plant DM02, receiving plant DM03
2	**Goods Issue — Supplying Plant**	MIGO — movement type 351, stock moves to in-transit
3	**Stock in Transit**	MMBE — in-transit stock visible at both plants during transfer
4	**Goods Receipt — Receiving Plant**	MIGO — movement type 101, stock updated at DM03 SL02

10.2 Consignment Process

6. Consignment PO created with item category K (ME21N)
7. GR posts to consignment stock type — no FI accounting document at receipt
8. Consignment stock visible in MBLB (vendor-owned stock at plant)
9. Withdrawal from consignment (movement type 201)—KON account key creates vendor payable
10. Periodic settlement via MRKO — consignment payable cleared to vendor account

10.3 Subcontracting — Pump Assembly

11. BOM created (CS01) for Pump Assembly (400001)—child components defined
12. Purchase Info Record created for subcontracting processing charges
13. Subcontracting PO with item category L (ME21N)
14. Components transferred to subcontractor: MIGO movement type 541 — special stock O
15. Finished Pump Assembly received: MIGO movement type 101 — components auto-consumed via 543

11. Batch Management and Split Valuation

11.1 Batch Management — Steel Sheet

Step	Activity	T-Code
1	**Activate Batch Management at Plant Level**	OMCE
2	**Create Batch Class (Class Type 023)**	CL01 — Class: DM_BATCH_CLASS
3	**Create Batch Characteristics**	CT04 — Characteristic: DM_GRADE | Values: Grade-A, Grade-B, Grade-C
4	**Assign Class to Material**	MM01 Classification view — class assigned to material 200001
5	**Create Batch Record at GR**	MSC1N — batch created with grade characteristic

11.2 Split Valuation — Steel Sheet by Quality Grade

Valuation Type	Category	Standard Price (EUR/KG)	GL Account (BSX)
GRADE-A	Quality Grade A — Premium	3.20	130001 Premium Steel
GRADE-B	Quality Grade B — Standard	2.50	130000 Standard Steel
GRADE-C	Quality Grade C — Economy	1.80	130002 Economy Steel

Split valuation activated via OMWC. Separate accounting views maintained per valuation type.

12. Physical Inventory

Full physical inventory cycle executed at plant DM01, storage location SL01.

Step	Activity	T-Code	Outcome
1	**Define PI Number Range**	OMBT	Number range assigned to PI document type
2	**Create PI Document**	MI01	PI document created for DM01/SL01 — all ROH materials included
3	**Enter Counted Quantities**	MI04	Physical count results entered per material line
4	**Review Differences**	MI20	Difference list: book quantity vs counted quantity
5	**Post Differences**	MI07	Stock adjusted, FI document via OBYC key INV

13. Testing and Validation

Test Area	Test Scenario	Expected Result
Pricing	Create PO 1,000 KG Steel Sheet from VEND-DM01	PB00: 2,500 | RA01: -125 | FRB1: 100 | Net: 2,475 EUR
Release Strategy	Create PO with value 250,000 EUR	PO blocked — all four release codes required
P2P Cycle	Execute full PR to MIRO and verify FI postings	BSX and WRX at GR | WRX cleared and vendor credited at MIRO

Split Valuation	GR for Grade-A and Grade-B batches separately	Separate stock quantities and separate GL accounts posted
Physical Inventory	Create PI document, count, post difference	Stock adjusted in MMBE | FI document via OBYC INV
ML Costing Run	Execute CKMLCP for period and review CKM3N	Actual cost differs from standard — revaluation entry posted

14. Project Summary

Configured complete three-plant enterprise structure for DMAG across Düsseldorf, Dortmund and Stuttgart
Built material master for ROH, HALB and FERT types with custom number ranges and valuation classes
Designed custom pricing procedure ZDM01 using condition technique with gross price, discount and freight
Implemented four-level PO release strategy using classification-based approval workflow
Activated Material Ledger with EUR and USD parallel currencies across all plants
Configured complete OBYC account determination covering BSX, WRX, PRD, GBB, FRE, KON and INV
Executed MRP Live with reorder point planning and PIR-based demand signals
Configured and tested STO, Consignment and Subcontracting with BOM
Implemented Batch Management with Split Valuation for Steel Sheet by quality grade
Executed complete Physical Inventory cycle with FI difference posting

PART 2

SAP S/4HANA MM — Procure-to-Pay (P2P) Cycle

PR → RFQ → Quotation → Comparison → PO → Goods Receipt → Invoice Verification

1. P2P Overview

The Procure-to-Pay cycle is the end-to-end process that begins when a business need is identified (Purchase Requisition) and ends when the vendor is paid (Invoice Verification). This document covers the full cycle executed for DMAG, including standard P2P, outline agreements, special scenarios and troubleshooting.

Scenario Context

Parameter	Value
Requesting Plant	DM02 — Dortmund Production
Purchase Organization	DMPO — Central Purchasing, DMAG
Material	200001 — Steel Sheet (ROH)
Quantity Required	1,000 KG
Competing Vendors	VEND-DM01 Steel Supplies GmbH | VEND-DM02 Aluminium Werke AG
Pricing Schema	ZDM01 — Gross Price + Discount + Freight

2. Step-by-Step P2P Execution

Step 1 — Purchase Requisition (PR)

PR raised by production planner for 1,000 KG Steel Sheet. Account assignment category K (cost centre DM02-PROD).

Configuration	Execution / Data
16. PR document type NB — number range, field selection configured 17. Account assignment categories K and P enabled	Material: 200001 Steel Sheet | Qty: 1,000 KG Plant: DM02 | Storage Loc: SL01 Acct Assignment: K — Cost Centre DM02-PROD **T-Code: ME51N**

18.	Mixed account assignment — cost split across cost objects	

Step 2 — Request for Quotation (RFQ)

Two RFQs created from PR via ME41 — one for each competing vendor.

Configuration		Execution / Data
19.	RFQ document type AN — dedicated number range configured	Vendor 1: VEND-DM01 Steel Supplies GmbH Vendor 2: VEND-DM02 Aluminium Werke AG Collective No: DM2026-001 | Deadline: 5 working days **T-Code: ME41**
20.	Pricing procedure ZDM01 active on quotation	
21.	Collective number assigned for multi-vendor comparison in ME49	

Step 3 — Maintain Quotation

Vendor prices entered via ME47. Pricing procedure ZDM01 calculates net effective price.

Vendor	Gross Price	Discount	Freight	Net Total (EUR)
VEND-DM01 Steel Supplies GmbH	2.50 EUR/KG	5% (>500 KG)	100 EUR flat	2,475.00
VEND-DM02 Aluminium Werke AG	2.70 EUR/KG	None	100 EUR flat	2,800.00

T-Code: ME47 | Validity: 30 days | Condition analysis visible on each quotation line item.

Step 4 — Quotation Comparison

ME49 price comparison executed for collective number DM2024-001. VEND-DM01 selected as lowest effective net price. VEND-DM02 marked for rejection.

22. VEND-DM01 selected: 2,475.00 EUR — saving 325.00 EUR vs VEND-DM02
23. Rejection reason code 01 applied to VEND-DM02 — rejection letter output triggered
24. T-Code: ME49

Step 5 — Purchase Info Record

25. Info Record created for VEND-DM01 and material 200001 at 2.50 EUR/KG
26. Planned delivery time: 10 days | Validity: 12 months
27. Delivery tolerance: 10% over, 10% under purchasing value key
28. Source List activated — VEND-DM01 set as fixed source for MRP auto-PO via ME59N
29. T-Code: ME11

Step 6 — Purchase Order (PO)

PO created for 1,000 KG Steel Sheet from VEND-DM01. Price auto-populated from Info Record. Net value 2,475 EUR — below 50,000 EUR release threshold.

PO Field	Value
Document Type	NB — Standard Purchase Order
Vendor	VEND-DM01 Steel Supplies GmbH
Material / Qty	200001 Steel Sheet | 1,000 KG | 2.50 EUR/KG
Net Value	2,475.00 EUR (gross 2,500 – discount 125 + freight 100)
Delivery	Plant DM02 | SL01 | Delivery date confirmed
GR-Based IV	Active — MIRO blocked until GR posted
Release Required	No — value 2,475 EUR below 50,000 EUR threshold
T-Code	ME21N

Step 7 — Goods Receipt (GR)

980 KG received (within 10% under-delivery tolerance). GR posted to DM02/SL01. Batch DM-BATCH-001 created with Grade-B classification.

GR Field	Value
Movement Type	101 — Goods Receipt for Purchase Order
Quantity	980 KG — within tolerance (1,000 KG ordered)
Stock Update	DM02 / SL01 Unrestricted Use: +980 KG
FI Posting Dr	130000 Inventory Account (BSX): 2,450.00 EUR
FI Posting Cr	191100 GR/IR Clearing (WRX): 2,450.00 EUR
T-Code	MIGO

Step 8 — Invoice Verification (MIRO)

Vendor invoice for 980 KG at 2.50 EUR/KG = 2,450 EUR. Three-way match: PO price, GR quantity and invoice amount all match. GR/IR clearing account cleared.

MIRO Field	Value
Invoice Amount	2,450.00 EUR | Vendor: VEND-DM01
Three-Way Match	PO price match | GR quantity match | Invoice amount match
FI Posting Dr	191100 GR/IR Clearing (WRX): 2,450.00 EUR — cleared
FI Posting Cr	Vendor Account VEND-DM01: 2,450.00 EUR — AP liability created
T-Code	MIRO

3. Additional P2P Scenarios

Scenario	Configuration	Key T-Codes
Value Contract	12-month framework, 500,000 EUR total. Item cat W (material group). Release orders drawn against contract.	ME31K / ME21N / ME3M
Scheduling Agreement	Periodic delivery for Aluminum Rod across 4 months. Type LP. Delivery schedule via ME38.	ME31L / ME38 / MIGO
Auto Purchase Order	Info Record and Source List active. ME59N converts approved PRs to POs automatically.	ME01 / ME11 / ME59N
Parked Invoice	Invoice saved without posting — pending approval. Reviewed and posted via MIR6.	MIRO / MIR6
GR Without PO	Unplanned receipt, movement type 501. Account assignment required at GR.	MIGO — 501
Unplanned Del Costs	Additional freight added at MIRO. Allocated to material cost or separate GL account.	MIRO
Quota Arrangement	10,000 KG split 60/40 between VEND-DM01 and VEND-DM02. Auto-PO assigns proportionally.	MEQ1 / ME01 / ME59N

4. L2/L3 Troubleshooting

Issue	Root Cause	Resolution Steps
GR/IR Mismatch	Invoice posted for different qty or value than GR. Or GR reversed	1. Run MB5S to identify open GR/IR items 2. Check PO history for GR and invoice quantities

	without corresponding invoice reversal.	3. 4.	Post credit memo in MIRO or reverse incorrect GR Clear residual items via MR11
PO Release Block	Release characteristic values on PO do not match configured strategy. Approver missing authorization.	5. 6. 7. 8.	Check release status in ME29N Verify CEKKO field values match strategy conditions Check user authorization for release code Reset release and re-trigger if classification incorrect
Pricing Error on PO	Wrong condition value on PO. Info Record price incorrect or access sequence not finding correct table.	9. 10. 11. 12.	Check condition analysis in PO (Analysis button) Verify Info Record price via ME13 Check access sequence ZDM1 for correct table hit Update condition record via MEK1 and re-determine price
MIRO Invoice Block	Invoice posted but payment blocked. Price or quantity variance outside tolerance. Manual block set.	13. 14. 15. 16.	Check tolerance key settings in OMR6 Review MIRO document for variance details Release block manually via MRBR if variance accepted Request credit memo from vendor for price discrepancy

5. FI-MM Integration Summary

Transaction	T-Code	Dr Account	Cr Account	OBYC Key
Goods Receipt	MIGO-101	130000 Inventory (BSX)	191100 GR/IR Clearing (WRX)	BSX/WRX
Invoice Posted	MIRO	191100 GR/IR Clearing (WRX)	Vendor Account (AP)	WRX
Price Difference	MIRO	281000 Price Variance (PRD)	Vendor Account (AP)	PRD
Goods Issue	MIGO-261	400000 Consumption (GBB-VBR)	130000 Inventory (BSX)	GBB/BSX
Return to Vendor	MIGO-122	191100 GR/IR Clearing (WRX)	130000 Inventory (BSX)	BSX/WRX

PART 3

SAP S/4HANA MM — Material Ledger and OBYC

Part 1 — Material Ledger

1.1 Overview

Material Ledger (ML) is mandatory in SAP S/4HANA. It serves two primary purposes:

17. Multi-currency inventory valuation: stock values recorded simultaneously in two parallel currencies (EUR and USD for DMAG)
18. Actual costing: calculates the true cost of goods at period end by accumulating all price variances from procurement and production

In S/4HANA, ML captures all variances throughout the period and distributes them automatically at period end via the CKMLCP costing run — giving finance teams actual cost visibility across all plants.

1.2 DMAG Material Ledger Design

Element	DMAG Setting	Rationale
ML Type	ZDM_ML — Actual Costing	Enables true cost accumulation across procurement and production
Plants Activated	DM01, DM02, DM03	All three plants require consistent actual costing for group reporting
Company Code Currency	EUR — Type 10	German statutory reporting currency
Group Currency	USD — Type 30	Parent company consolidation currency
Price Control (ROH/HALB)	V — Moving Average Price	Reflects actual market price fluctuations for raw materials
Price Control (FERT)	S — Standard Price	Enables variance analysis between planned and actual production cost
Costing Run Frequency	Monthly — CKMLCP	Period-end actual cost determination and inventory revaluation

1.3 Activation Steps

Step	T-Code	Activity	DMAG Configuration
1	**OMX2**	Define Valuation Areas	Valuation areas defined for plants DM01, DM02, DM03
2	**OMX3**	Assign ML Type to Valuation Area	ZDM_ML assigned to all three valuation areas
3	**CKMSTART**	Activate and Initialize Material Ledger	ML activated — status PRODUCTIVE. Opening balances set.
4	**OBY6**	Configure Currency Types	Type 10: EUR (Company Code) | Type 30: USD (Group)
5	**OMRW**	Set Valuation Method	Price determination 3 — Single and Multi-Level Actual Costing
6	**CKMLCP**	Execute Actual Costing Run	Period-end run — actual cost calculated and inventory revalued
7	**CKM3N**	Material Price Analysis	Review actual vs standard cost per material, per plant, per period

SAP System Evidence *T-Code: CKMSTART / CKM3N*	
Navigation: *SPRO → Controlling → Product Cost Controlling → Actual Costing/Material Ledger → Activate Valuation Areas*	
ML Status:	`DM01 PRODUCTIVE | DM02 PRODUCTIVE | DM03 PRODUCTIVE`
ML Type:	`ZDM_ML | Price Determination: 3 (Actual Costing)`

Currencies:	`Type 10: EUR | Type 30: USD`
Steel Sheet:	`Standard 2.40 EUR/KG | Actual 2.51 EUR/KG after CKMLCP`
Machined Housing:	`Standard 3.20 EUR/PC | Actual 3.30 EUR/PC after CKMLCP`
What to verify: *All three plants show PRODUCTIVE status. CKM3N shows parallel EUR and USD values with actual cost after costing run.*	

1.4 Actual Costing Scenario — Steel Sheet to Machined Housing

This scenario demonstrates how Material Ledger accumulates actual costs from procurement through production.

Stage	Activity	EUR Cost	USD Parallel	ML Impact
1. Procurement	Steel Sheet purchased at 2.50 EUR/KG — standard price 2.40 EUR/KG	2,500 EUR	2,720 USD	Variance 100 EUR captured by ML
2. Production Issue	Steel Sheet issued to production order for Machined Housing	2,500 EUR actual	2,720 USD	Actual material cost to WIP
3. Processing	Machining activity confirmed on production order: 0.80 EUR/PC	800 EUR	870 USD	Activity cost added to order
4. GR to Stock	Machined Housing received to stock at standard price 3.20 EUR/PC	3,200 EUR std	3,480 USD	Variance 100 EUR held in ML
5. Period End	CKMLCP run — variances rolled up, inventory revalued to actual cost	3.30 EUR/PC actual	3.59 USD/PC	Revaluation posted via OBYC

1.5 Multi-Currency Valuation

Currency Type	Currency	Exchange Rate	Example: 1,000 KG Steel Sheet
10 — Company Code Currency	EUR	Base currency	2,500.00 EUR
30 — Group Currency	USD	1 EUR = 1.088 USD	2,720.00 USD

Exchange rates maintained in OB08. Applied automatically at each posting. Both currencies are visible in all ML reports and FI documents.

Part 2 — OBYC: Automatic Account Determination

2.1 Overview

OBYC (Configure Automatic Postings) maps SAP MM goods movements to Financial Accounting GL accounts. Without OBYC, every goods movement would require manual account selection. The transaction key, valuation class and account grouping determine which GL account is posted automatically.

The valuation class on the material master is the primary driver for BSX (inventory) postings. Different material types post to different inventory GL accounts.

2.2 DMAG GL Account Structure

GL Account	Account Name	OBYC Key	Valuation Class / Trigger
130000	Raw Material Inventory — Steel Sheet, Aluminum Rod	BSX	3000 — Raw Material
131000	Semi-Finished Goods — Machined Housing	BSX	7920 — Semi-Finished
132000	Finished Goods — Pump Assembly	BSX	7900 — Finished Goods

191100	GR/IR Clearing Account	WRX	All material types — both sides of GR and MIRO
281000	Purchase Price Variance	PRD	Standard price materials — invoice vs PO difference
400000	Material Consumption — Production	GBB-VBR	GI to production order (261)
401000	Material Consumption — Cost Centre	GBB-VBO	GI to cost center (201)
410000	Freight Clearing Account	FRE	Freight condition at GR posting
160000	Consignment Payable — Vendor	KON	Withdrawal from consignment stock (201)
170000	Inventory Difference Account	INV	Physical inventory difference posting (MI07)

2.3 Transaction Key Detail

BSX — Inventory Posting

Triggered by every goods movement that changes stock quantity and value: GR (101), GI (261), STO (351/101), return to vendor (122), transfer posting (301). Account selected by valuation class from material master.

Movement	Dr Account	Cr Account
GR (101)	130000 Raw Material Inventory	191100 GR/IR Clearing (WRX)
GI to Production (261)	400000 Consumption-Production (GBB-VBR)	130000 Raw Material Inventory
Return to Vendor (122)	191100 GR/IR Clearing (WRX)	130000 Raw Material Inventory

WRX — GR/IR Clearing Account

The GR/IR account (191100) is a bridge between GR and MIRO. It should net to zero when GR and invoice quantities match. Open items on WRX indicate unmatched GRs or invoices.

Event	WRX Posting	Net Result
At GR (101)	Cr 191100 GR/IR Clearing: 2,450.00 EUR	Open item created — liability to vendor
At MIRO	Dr 191100 GR/IR Clearing: 2,450.00 EUR	Open item cleared — vendor account credited
Net (matched)	191100 balance = 0.00 EUR	Accounting cycle complete for this delivery

PRD — Price Differences

Triggered when invoice price differs from PO price for standard price (S) materials. For Pump Assembly FERT (standard price 8.50 EUR/PC, invoice 8.70 EUR/PC):

Posting Event	Dr	Cr
GR posting	132000 Finished Goods (BSX): 8,500.00	191100 GR/IR Clearing (WRX): 8,500.00
MIRO posting	191100 GR/IR: 8,500 | PRD 281000: 200	Vendor Account: 8,700.00

GBB — Offsetting Entries for Goods Issues

GBB account selection uses both the account modification key (VBR, VBO etc.) and the valuation class. Allows different consumption accounts per material type and purpose.

Sub-Key	Movement	Dr Account	Cr Account
GBB-VBR	GI to Production (261)	400000 Consumption-Production	130000 Inventory (BSX)
GBB-VBO	GI to Cost Centre (201)	401000 Consumption-Cost Centre	130000 Inventory (BSX)
GBB-BSA	Initial Stock Entry (561)	130000 Inventory (BSX)	Offsetting account

KON — Consignment Payable

Unlike standard stock, consignment GR creates no FI posting — stock is recorded but vendor retains ownership. Liability only arises at consumption.

19. Consignment GR (101): No FI document — stock recorded in consignment stock type
20. Consignment GI (201): Dr 400000 Consumption | Cr 160000 Consignment Payable (KON)
21. Periodic Settlement (MRKO): 160000 Consignment Payable cleared to vendor account for payment

SAP System Evidence *T-Code: OBYC*

Navigation: *SPRO → Materials Management → Valuation and Account Assignment → Account Determination Without Wizard → Configure Automatic Postings*

BSX Val 3000:	→ GL 130000 Raw Material Inventory
BSX Val 7920:	→ GL 131000 Semi-Finished Inventory
BSX Val 7900:	→ GL 132000 Finished Goods Inventory
WRX (all mats):	→ GL 191100 GR/IR Clearing Account
PRD (std price):	→ GL 281000 Purchase Price Variance
GBB-VBR:	→ GL 400000 Consumption Production
KON (all mats):	→ GL 160000 Consignment Payable
INV (all mats):	→ GL 170000 Inventory Difference

What to verify: *Transaction key BSX showing valuation class to GL mapping. WRX single account 191100 for all materials. PRD to 281000. GBB sub-keys with separate consumption accounts.*

Part 3 — Period-End Closing Sequence

Step	T-Code	Activity	ML / OBYC Impact
1	**MB5S**	GR/IR Clearing Review	Identify all open GR/IR items — match uncleared GRs with missing invoices
2	**MMPV**	MM Period Close	Close posting period — prevent backdated postings to previous period
3	**KO88**	Production Order Settlement	Settle production order variances to cost of goods manufactured — feeds ML
4	**CKMLCP (test)**	Preliminary Costing Run	Test run — preview actual cost calculations without posting
5	**CKMLCP (final)**	Actual Costing Run	ML calculates actual cost. Posts revaluation entries via OBYC BSX and PRD.
6	**CKM3N**	Material Price Analysis	Verify actual cost per material, per plant. Review variance layers and cost component split.

SAP System Evidence *T-Code: CKMLCP*

Navigation: *SAP Easy Access → Accounting → Controlling → Product Cost Controlling → Actual Costing → Edit Costing Run*

Costing Run:	DM_OCT2025 | Period: 10/2025 | Company Code: DM01
Steps Completed:	Selection → Sequence → Single-Level → Multi-Level → Revaluation → Post Closing
Steel Sheet:	Actual 2.51 EUR/KG vs Standard 2.40 | Variance 110 EUR revalued
Machined Housing:	Actual 3.30 EUR/PC vs Standard 3.20 | Variance 100 EUR revalued
FI Impact:	Revaluation posted to BSX and PRD accounts — inventory restated to actual cost

What to verify: *CKMLCP log shows all six steps completed without errors. CKM3N confirms actual price update for all materials in all plants.*

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